SINGAPORE SCHOOL KINDERLAND

Department of Information Technology & Computer Science · Academic Year 2026–2027

CAMBRIDGE O LEVEL COMPUTER SCIENCE (0478/2210) · CLASS TEST 1 (TERM 2) · MARK SCHEME

Official Evaluation Criteria & Examiner Guidance · Total: 25 Marks

Document: Cambridge Standard Mark Scheme	**Subject / Code:** Computer Science (0478/2210)	**Maximum Mark:** 25

SECTION A: TRANSMISSION MECHANICS & MULTIPLE CHOICE [5 MARKS]

1. **(c) Full-duplex** (Both directions at same time) [1 Mark]

2. **(b) Serial** (1 bit at a time sequentially) [1 Mark]

3. **(b) Bit skew** (Desynchronization of bits across multiple wires) [1 Mark]

4. **(c) Packet Header** (Contains routing metadata, sequence number, IP addresses) [1 Mark]

5. **(c) USB** (Universal Serial Bus) [1 Mark]

SECTION B: SHORT STRUCTURED QUESTIONS [6 MARKS]

1. **Packet Components:** [3 Marks]
• Packet Header (Source IP, Destination IP, Packet Sequence Number, Packet Size)
• Payload (The actual chunk of user data being transmitted)
• Packet Trailer / Footer (Checksum/CRC error-detection code, end-of-packet marker)

2. **Routing & Reassembly:** [3 Marks]
• Routers assess real-time network congestion and hop availability, sending independent packets along whichever route is fastest at that instant [1.5 Marks].
• The receiving device checks packet sequence numbers in the header and reorders them into the correct original order [1.5 Marks].

SECTION C: DIRECTIONAL TRANSMISSION [4 MARKS]

• Wireless Mouse to USB Dongle: **Simplex** [1 Mark]
• Two-way Video Call (Zoom): **Full-duplex** [1 Mark]
• Walkie-talkies: **Half-duplex** [1 Mark]
• Digital TV Broadcast: **Simplex** [1 Mark]

SECTION D: SERIAL VS PARALLEL TRANSMISSION ANALYSIS [5 MARKS]

1. Comparison Table: [3 Marks]

• Number of data lines: Serial = 1 single wire/channel; Parallel = Multiple wires (e.g. 8, 16, 32, 64) [1 Mark]

• Risk of bit skew / crosstalk: Serial = No bit skew, low crosstalk; Parallel = High risk of bit skew & electrical interference [1 Mark]

• Operational distance: Serial = Long distances (up to thousands of km); Parallel = Short distances (within a few meters/on a PCB) [1 Mark]

2. Why Serial Replaced Parallel: [2 Marks]

• Elimination of bit skew allows serial connections to run at much higher clock frequencies (Gigabits/sec).

• Thinner, cheaper cables with fewer wires improve airflow inside computer chassis and reduce production costs.

SECTION E: UNIVERSAL SERIAL BUS (USB) [5 MARKS]

1. **Benefits:** [3 Marks]

• Automatically detects and loads drivers (Plug and Play).

• Provides low-voltage DC power (5V) to charge or run devices without separate power bricks.

• Universal industry standard compatible with virtually all modern computers and operating systems.

• Connectors only fit one way (Type-A) or are reversible (Type-C), preventing incorrect cable insertion.

2. **Limitation:** [1 Mark]

Maximum cable length without active repeaters is limited (typically 5 meters max for USB 2.0 / 3 meters for USB 3.0).

3. **Plug-and-Play Detection:** [1 Mark]

When plugged in, the device causes a small voltage change on the D+/D- data lines; the OS detects this and queries the device descriptor to load the matching driver.